

SQL-Based Data Analysis Using Superstore Dataset

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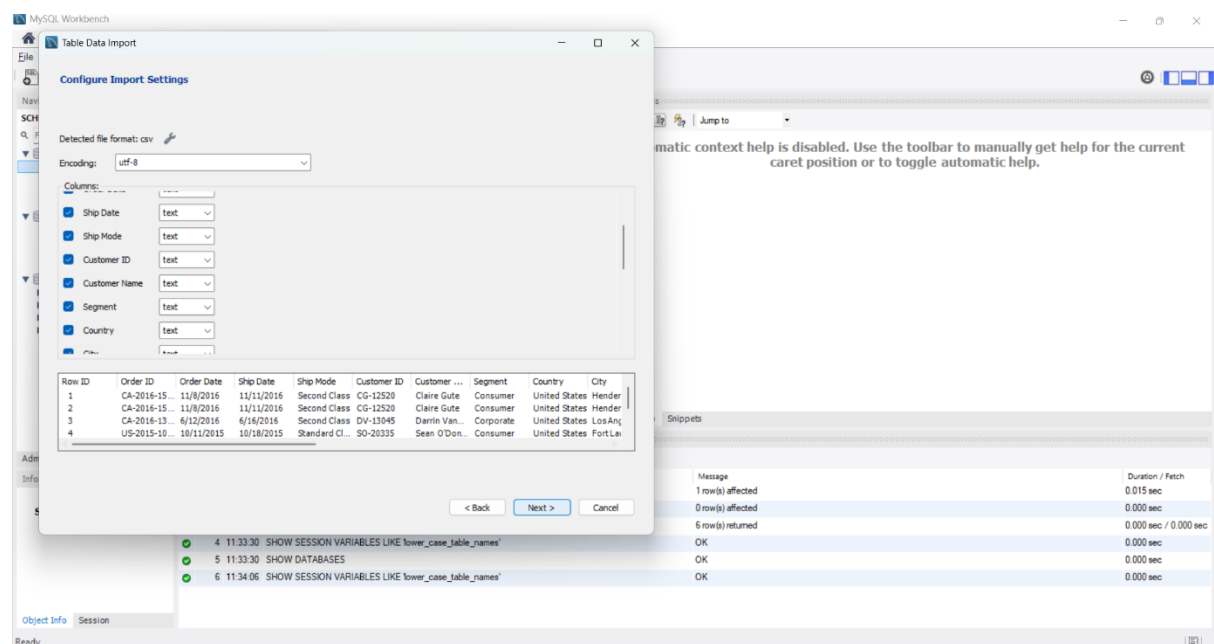
Objective :

Analyze sales data using SQL with filtering, aggregation, and business queries. Steps: 1.Load dataset into a SQL database. 2.Explore table (schema, sample data). 3.Apply WHERE filters (region, category, date, sales). 4.Use GROUP BY for aggregations (sales, quantity, averages). 5.Sort and limit results (top products, top categories). 6.Solve use cases (monthly trends, top customers, duplicates). 7.Validate results (row counts, data quality). Output: SQL script / Notebook + query results + brief insights.

Resources : <https://www.kaggle.com/datasets/vivek468/superstore-dataset->

1. Load Dataset into SQL Database

The Superstore dataset was imported into MySQL Workbench and stored in a table named superstore.

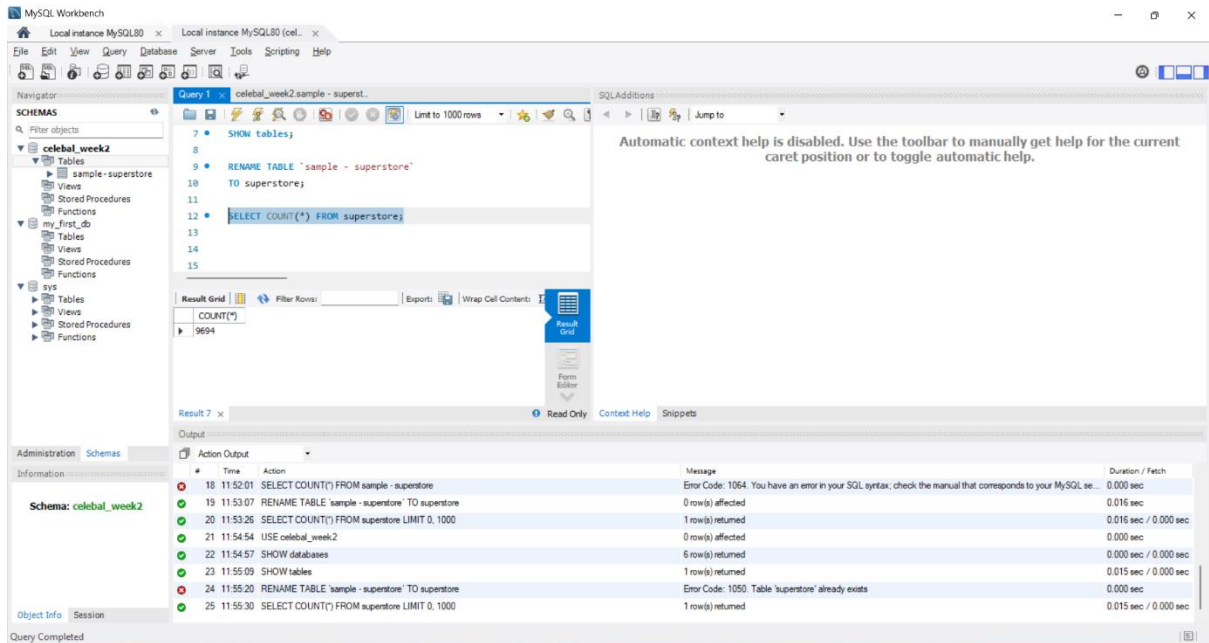


Query

```
SELECT COUNT(*) FROM superstore;
```

Result

Total Records Loaded: 9694



Insight

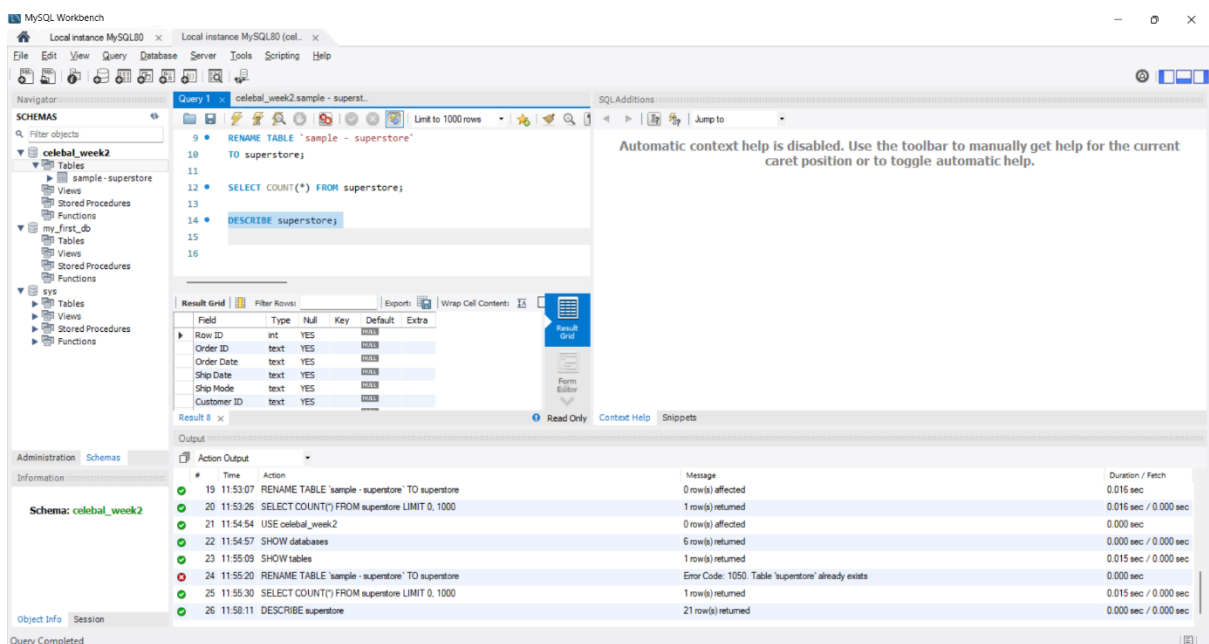
The Superstore dataset was successfully imported into MySQL Workbench. The row count verification confirmed that all records were loaded correctly and the dataset was ready for analysis.

2. Explore Table (Schema and Sample Data)

To understand the dataset structure, the schema and sample records were examined.

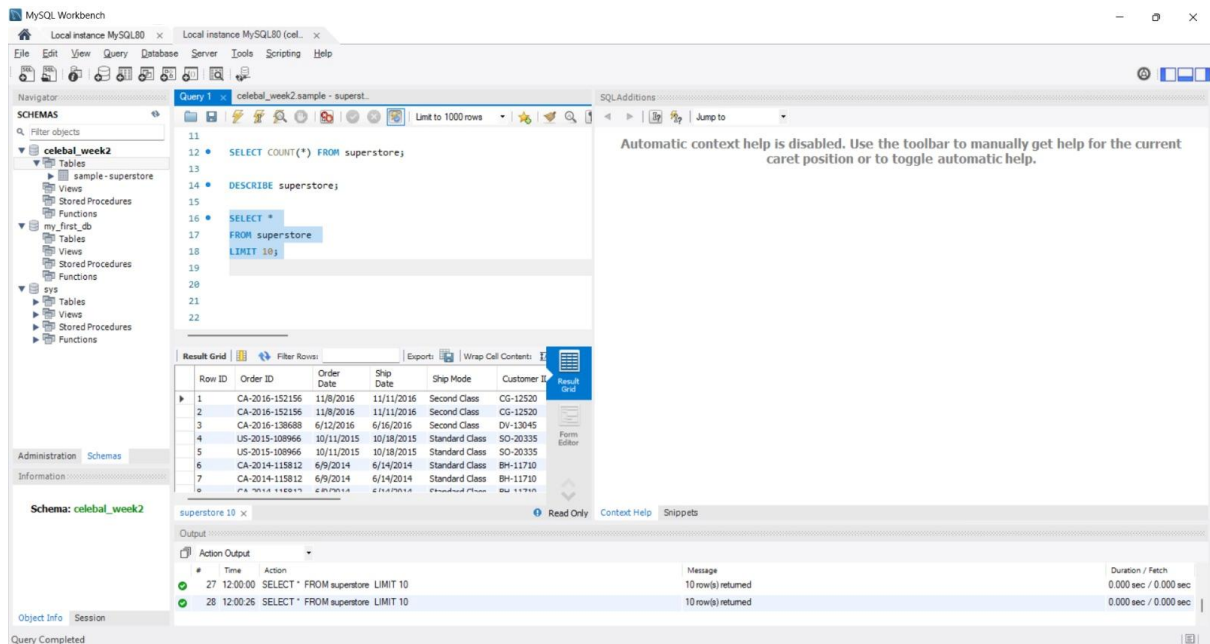
Query 1

DESCRIBE superstore;



Query 2

```
SELECT *  
FROM superstore  
LIMIT 10;
```



Insight

Exploring the table structure helped in understanding the available columns, their data types, and the overall organization of the dataset. Reviewing sample records provided a clear idea of the information available for analysis

3. Apply WHERE Filters (Region, Category, Sales)

Filtering was performed to retrieve specific subsets of data.

Region Filter

```
SELECT *  
FROM superstore  
WHERE Region = 'South';
```

Category Filter

```
SELECT *  
FROM superstore  
WHERE Category = 'Technology';
```

Sales Filter

SELECT *
FROM superstore
WHERE Sales > 1000;

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```

76 ORDER BY total_profit DESC;
77
78 -- top 5 loss making products
79 SELECT "Product Name",
80 SUM(Profit) AS total_profit
81 FROM superstore
82 GROUP BY "Product Name"
83 ORDER BY total_profit ASC
84 LIMIT 5;
85
86 -- where filter for region
87 SELECT *
88 FROM superstore
89 WHERE Region = 'South';

```

The result grid displays a list of products and their total profit, sorted by profit in descending order. The first five rows are highlighted in blue.

Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
Jane Gulte	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases
Jane Gulte	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs
Jan O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables
Jan O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage
Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Paper

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```

87 SELECT *
88 FROM superstore
89 WHERE Region = 'South';
90
91 -- where filter for category
92 SELECT *
93 FROM superstore
94 WHERE Category = 'Technology';
95
96 -- where filter for sales
97 SELECT *
98 FROM superstore
99 WHERE Sales > 1000;
100

```

The result grid displays a list of products and their sales, sorted by sales in descending order. The first five rows are highlighted in blue.

Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092
TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	1044.63	3	0	240.2649
BO-10004834	Furniture	Bookcases	Riverside Palms Royal Lawyers Bookcase, Royal...	3083.43	7	0.5	-1665.0522
PH-10004977	Technology	Phones	GE 30524E24	1097.544	7	0.2	125.4737
PH-10002447	Technology	Phones	AT&T CL83451 4-Handset Telephone	1028.95	5	0	288.6855

Insight

Filters were applied to focus on specific subsets of data. This helped analyze sales records from particular regions, product categories, and high-value transactions more effectively.

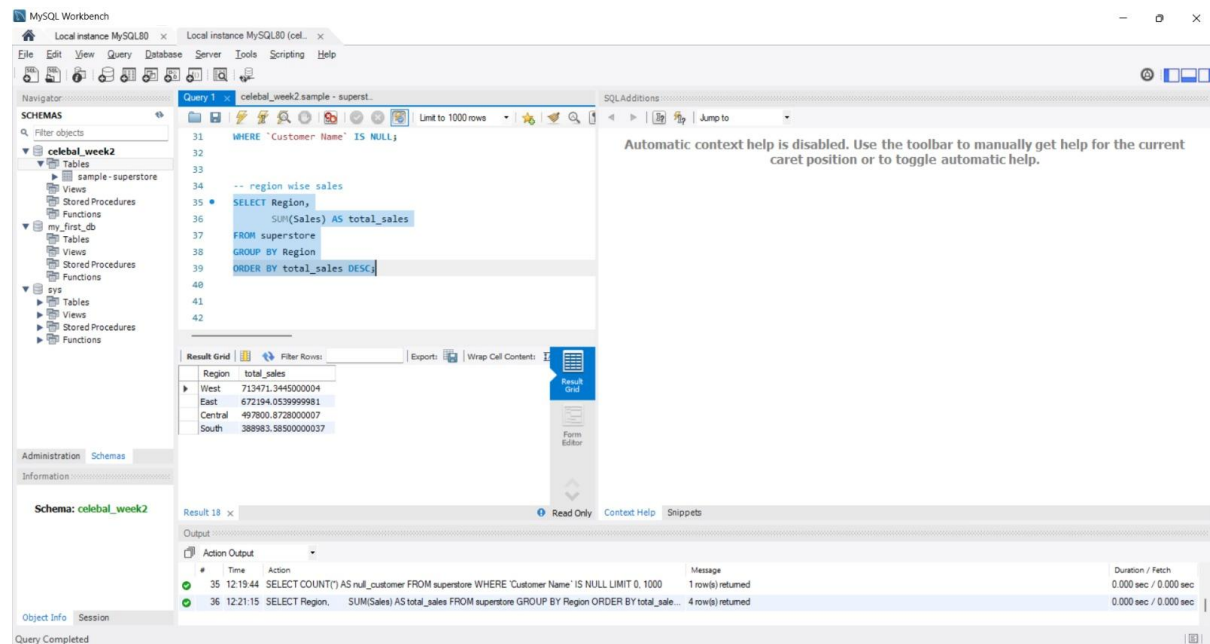
4. Use GROUP BY for Aggregations

Aggregations were performed using GROUP BY to summarize sales and profit information.

Region-Wise Sales

```
SELECT Region,  
SUM(Sales) AS total_sales  
FROM superstore  
GROUP BY Region  
ORDER BY total_sales DESC;
```

Output :



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
31 WHERE 'Customer Name' IS NULL;  
32  
33  
34 -- region wise sales  
35 * SELECT Region,  
36 SUM(Sales) AS total_sales  
37 FROM superstore  
38 GROUP BY Region  
39 ORDER BY total_sales DESC;  
40  
41  
42
```

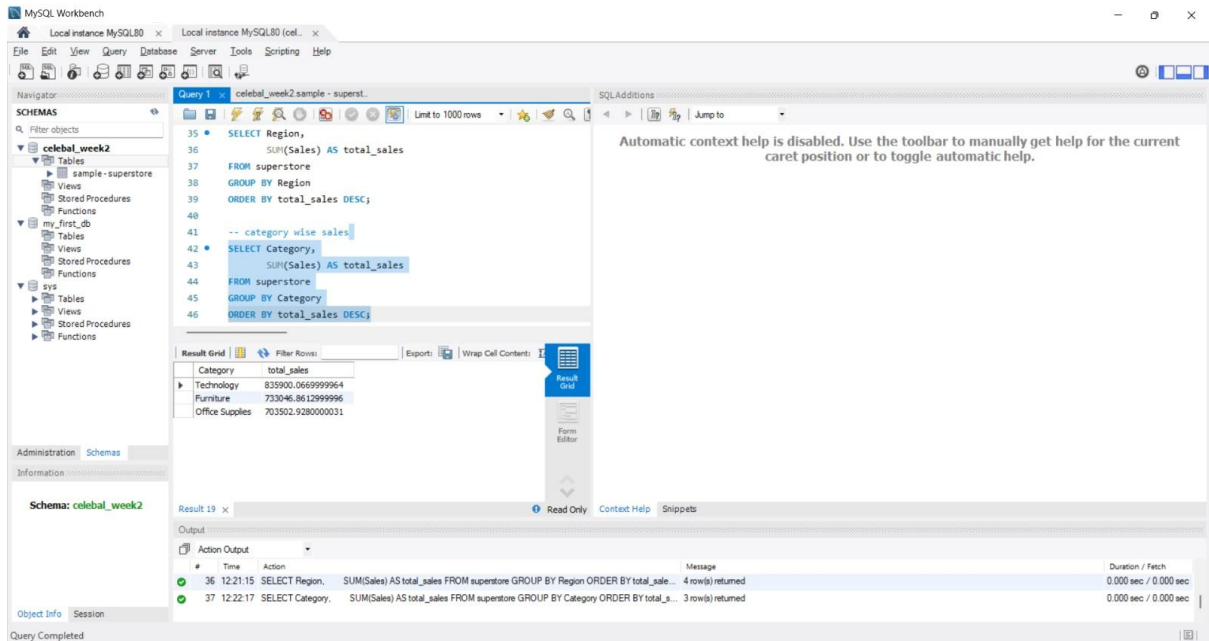
The Results window displays the output of the query:

Region	total_sales
West	713471.3445000004
East	672194.0539999981
Central	497800.8728000007
South	388983.58500000037

The bottom status bar indicates "Query Completed".

Category-Wise Sales

```
SELECT Category,  
SUM(Sales) AS total_sales  
FROM superstore  
GROUP BY Category  
ORDER BY total_sales DESC;
```

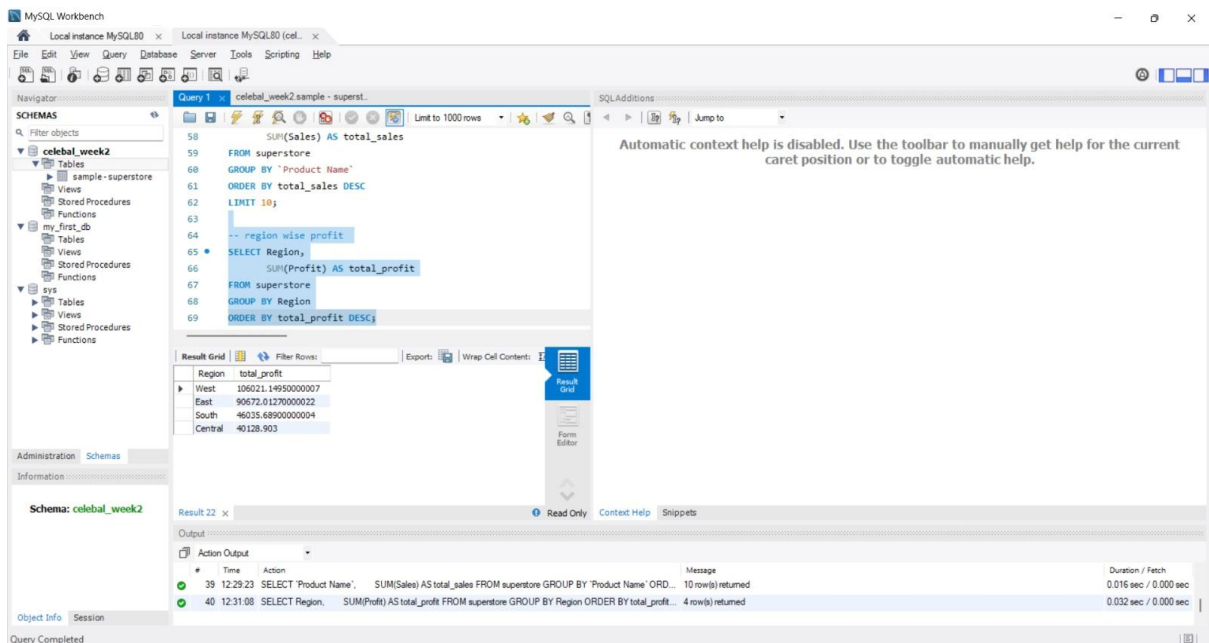


Region-Wise Profit

```

SELECT Region,
SUM(Profit) AS total_profit
FROM superstore
GROUP BY Region
ORDER BY total_profit DESC;

```



Category-Wise Profit

```

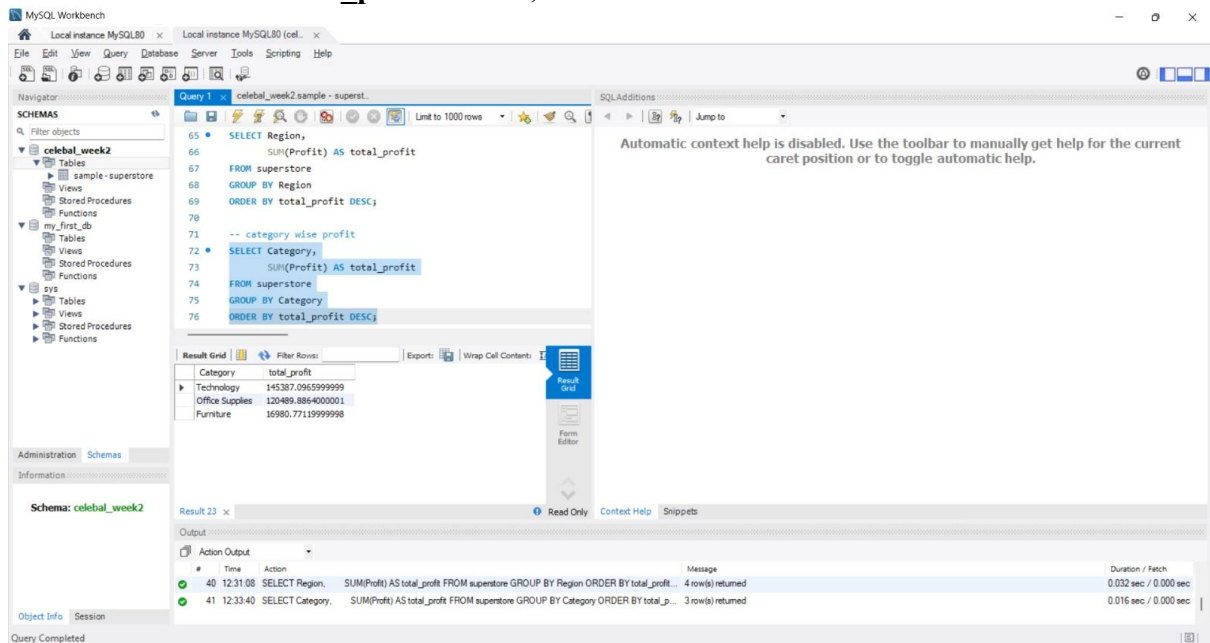
SELECT Category,
SUM(Profit) AS total_profit

```


FROM superstore

GROUP BY Category

ORDER BY total_profit DESC;



Insight

Aggregation queries summarized sales and profit values across different regions and categories. This made it easier to compare performance and identify key contributors to business revenue.

5. Sort and Limit Results (Top Products and Customers)

Sorting and limiting operations were used to identify top-performing entities.

Top 10 Customers

```

SELECT `Customer Name`,
SUM(Sales) AS total_sales
FROM superstore
GROUP BY `Customer Name`
ORDER BY total_sales DESC
LIMIT 10;

```

MySQL Workbench interface showing a query executed on the 'celestial_week2' database. The query is as follows:

```

43  SUM(Sales) AS total_sales
44  FROM superstore
45  GROUP BY Category
46  ORDER BY total_sales DESC;
47
48  -- top 10 customers
49  * SELECT 'Customer Name',
50      SUM(Sales) AS total_sales
51  FROM superstore
52  GROUP BY 'Customer Name'
53  ORDER BY total_sales DESC
54  LIMIT 10;

```

The result grid displays the top 10 customers by total sales:

Customer Name	total_sales
Sean Miller	25043.05
Tamara Chand	19017.847999999998
Raymond Buch	15117.339
Tom Ashbrook	14995.62
Adrian Barton	14355.610999999997
Sargit Chand	14142.339999999999
Ken Lonsdale	14071.917
Hunter Lopez	12873.297999999999
Sargit Engle	12209.438000000002

The output section shows the execution details for the query:

#	Time	Action	Message	Duration / Fetch
37	12:23:17	SELECT Category, SUM(Sales) AS total_sales FROM superstore GROUP BY Category ORDER BY total_s...	3 row(s) returned	0.000 sec / 0.000 sec
38	12:23:10	SELECT 'Customer Name', SUM(Sales) AS total_sales FROM superstore GROUP BY 'Customer Name' O...	10 row(s) returned	0.016 sec / 0.000 sec

Top 10 Products

```

SELECT 'Product Name',
SUM(Sales) AS total_sales
FROM superstore
GROUP BY 'Product Name'
ORDER BY total_sales DESC
LIMIT 10;

```

MySQL Workbench interface showing a query executed on the 'celestial_week2' database. The query is as follows:

```

51  FROM superstore
52  GROUP BY 'Customer Name'
53  ORDER BY total_sales DESC
54  LIMIT 10;
55
56  -- top 10 products
57  * SELECT 'Product Name',
58      SUM(Sales) AS total_sales
59  FROM superstore
60  GROUP BY 'Product Name'
61  ORDER BY total_sales DESC
62  LIMIT 10;

```

The result grid displays the top 10 products by total sales:

Product Name	total_sales
Canon imageCLASS 2200 Advanced Copier	61599.824
Pellicore PR500 Electric Punch Plastic Comb Bind...	27453.384
Cisco TelePresence System EX90 Videoconferen...	22638.40
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479000000003
GBC iGenMaster 500 Manual ProClick Binding System	19024.5
Hewlett Packard Laser Jet 3330 Copier	18839.686
HP DesignJet T530 Inkjet Large Format Printer ~...	18374.895
GBC DocuBind P400 Electric Binding System	17965.968

The output section shows the execution details for the query:

#	Time	Action	Message	Duration / Fetch
38	12:23:10	SELECT 'Customer Name', SUM(Sales) AS total_sales FROM superstore GROUP BY 'Customer Name' O...	10 row(s) returned	0.016 sec / 0.000 sec
39	12:23:23	SELECT 'Product Name', SUM(Sales) AS total_sales FROM superstore GROUP BY 'Product Name' ORD...	10 row(s) returned	0.016 sec / 0.000 sec

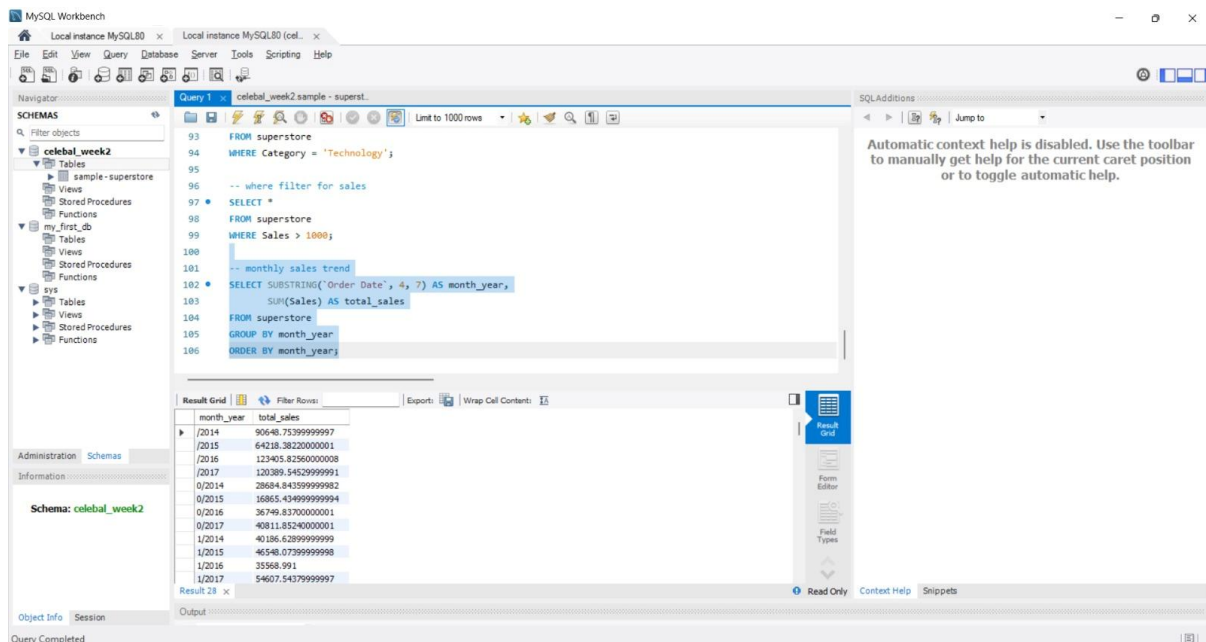
Insight

Sorting and limiting the results helped identify the most valuable customers and best-performing products based on total sales. These insights can support business decision-making and customer targeting.

6. Solve Business Use Cases

Monthly Sales Trend

```
SELECT SUBSTRING('Order Date',4,7) AS month_year,  
SUM(Sales) AS total_sales  
FROM superstore  
GROUP BY month_year  
ORDER BY month_year;
```

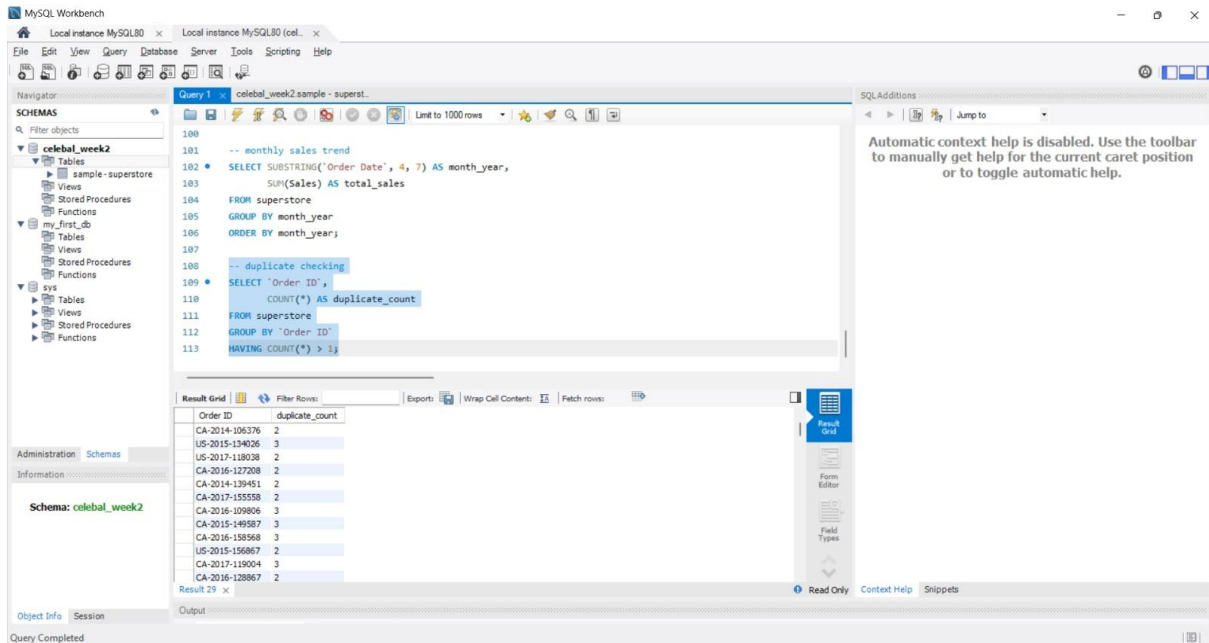


Insight

The monthly sales trend analysis provided an overview of how sales changed over time. Such analysis can help identify seasonal patterns and periods of strong or weak performance.

Duplicate Order Analysis

```
SELECT Order ID,  
COUNT(*) AS duplicate_count  
FROM superstore  
GROUP BY Order ID  
HAVING COUNT(*) > 1;
```



Insight

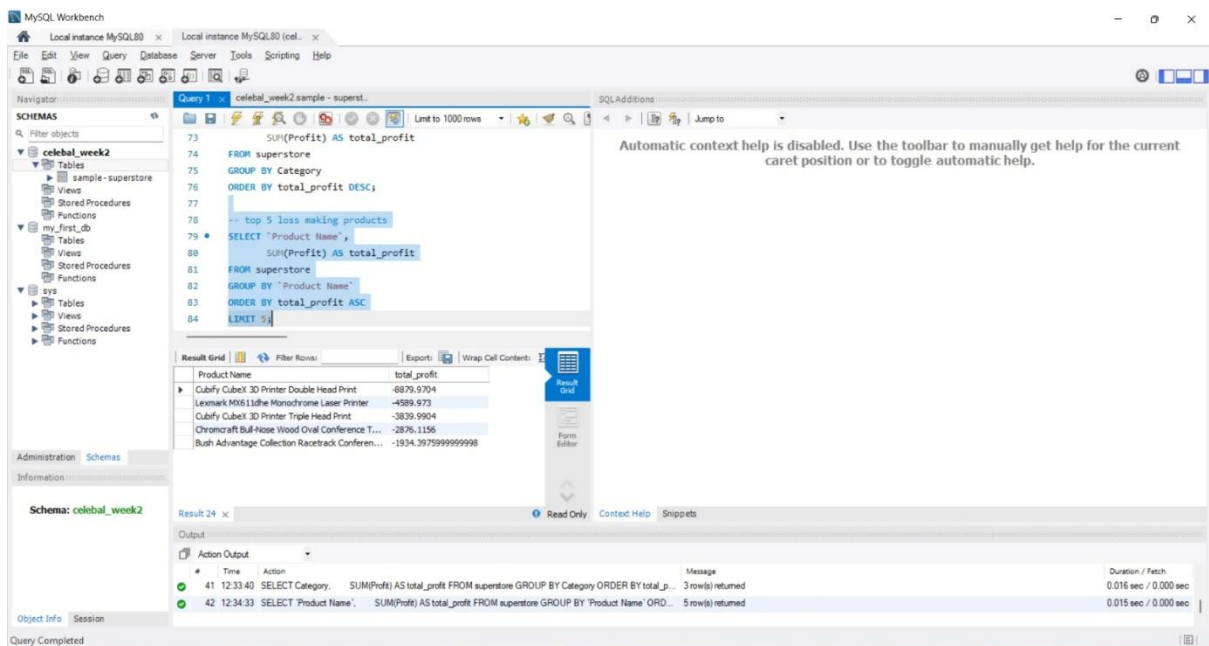
Duplicate order analysis was performed to check the consistency of order records and identify any repeated entries that may affect reporting accuracy.

Loss-Making Products

```

SELECT 'Product Name',
SUM(Profit) AS total_profit
FROM superstore
GROUP BY 'Product Name'
ORDER BY total_profit ASC
LIMIT 5;

```



Insight

The analysis highlighted products generating the lowest profits. Identifying such products can help businesses review pricing strategies and improve profitability.

7. Validate Results (Row Counts and Data Quality)

Data validation was performed to ensure data completeness and reliability.

Query 1

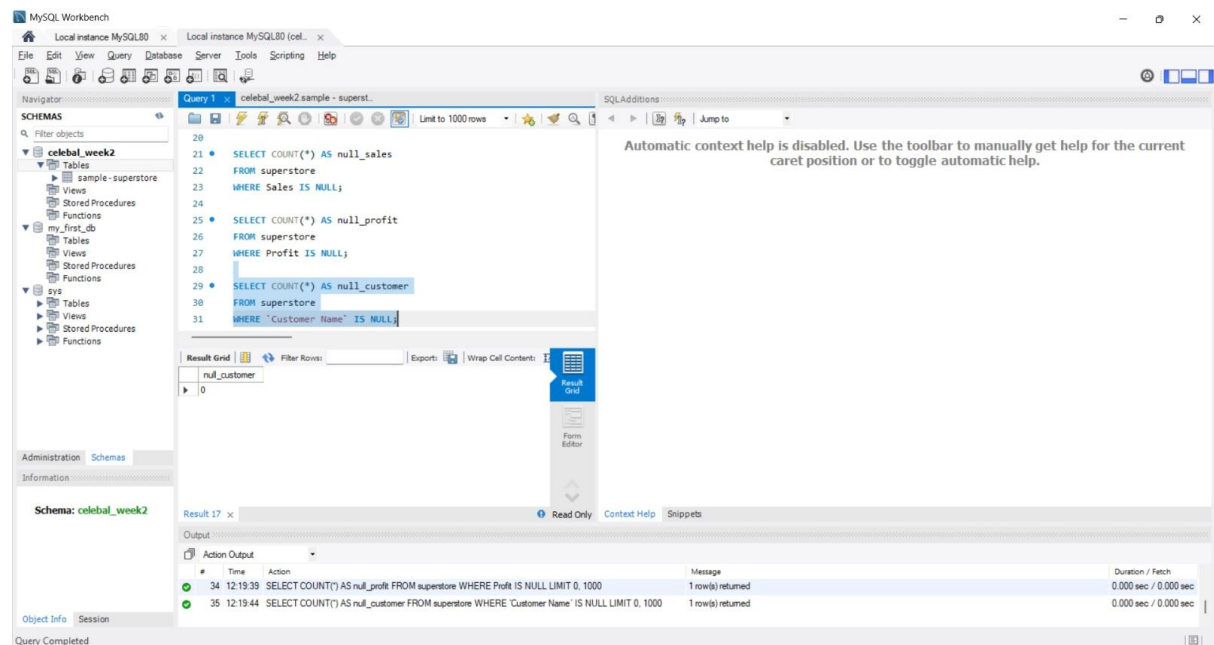
```
SELECT COUNT(*) AS null_sales
FROM superstore
WHERE Sales IS NULL;
```

Query 2

```
SELECT COUNT(*) AS null_profit
FROM superstore
WHERE Profit IS NULL;
```

Query 3

```
SELECT COUNT(*) AS null_customer
FROM superstore
WHERE `Customer Name` IS NULL;
```



Result

NULL Sales = 0

NULL Profit = 0

NULL Customer Name = 0

Insight

Data quality checks showed that important business fields such as Sales, Profit, and Customer Name did not contain missing values. This indicates that the dataset is reliable for further analysis.

Overall Findings :

- Successfully imported and analyzed 9,694 sales records.
- No missing values were found in major business columns.
- Sales and profit performance were analyzed across regions and categories.
- Top customers and top products were identified.
- Monthly sales trends were evaluated.
- Duplicate records and data quality checks were performed.
- Loss-making products were identified for business review.

Conclusion :

The Superstore dataset was successfully analyzed using SQL in MySQL Workbench. Various SQL concepts including filtering, aggregation, sorting, grouping, and data validation were applied to answer business-related questions. The analysis helped identify top-performing customers and products, evaluate regional and category-wise performance, examine monthly sales trends, and verify data quality. Overall, the project demonstrated how SQL can be used to transform raw sales data into meaningful business insights.